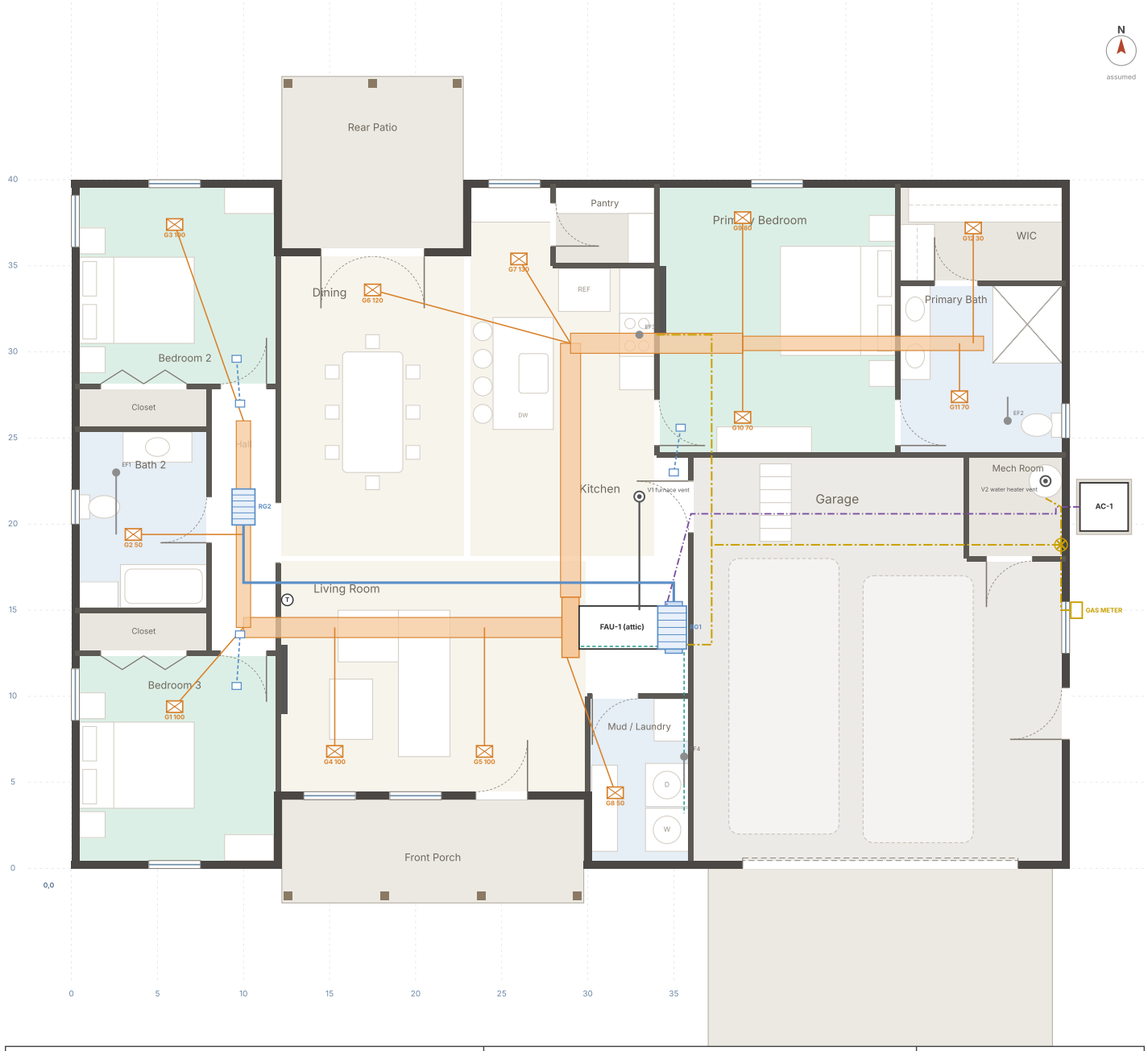


Waring Build BuildWise plan 3933 · 3 bed, 2 bath · 1,800 sq ft + 528 sq ft garage · 58'-0" x 40'-0" framing Slab on grade · 9' ceilings · 6:12 gable · grid 5', 0,0 at the south-west corner	**HVAC plan** Gas furnace FAU-1 in the attic, A/C AC-1, gas piping, ducts, registers, exhaust Schematic for pricing and coordination; equipment and duct sizes need a Manual J/S/D.	**M-1** rev drawn for K. Christianson

M-1 HVAC

supply trunks and flex, ceiling diffusers G# returns RG# and jump ducts J# refrigerant line set condensate natural gas appliance vents V# exhaust ducts EF#

Notes

1. The gas furnace FAU-1 sits in the attic on a platform over the back hall, as the owner asked; the mechanical room is in the garage (water heater, panel, manifold). The way to FAU-1 is the pull-down stair in the garage ceiling, with a catwalk and a 30" service space (M1305.1.3), a light switched at the access and a receptacle at the unit.
2. Every duct is in the attic: sheet-metal trunks with R-8 wrap, R-8 flex to ceiling diffusers, every joint sealed with mastic. Keep the flex runs short and straight, and pull them tight. Bury the ducts in the attic insulation where they cross the ceiling.
3. Airflow (1,000 CFM) is shared by room area and exposure for pricing; the HVAC contractor's Manual J (loads), S (equipment) and D (ducts) set the final numbers. Two filtered returns: RG1 under the unit and RG2 in the bedroom hall; the bedrooms return through jump ducts J1–J3.
4. Heat: natural gas. FAU-1 is a sealed-combustion condensing furnace (direct vent), so it takes no combustion air from the attic; its 2" PVC intake and exhaust leave through the rear roof slope (V1). Cooling from AC-1 through the coil on the furnace. Zone 6 assumed; confirm the design temperature.
5. Gas: the meter is on the garage east wall. The line runs up the inside of that wall into the mech room (main shutoff, capped tee for later), up into the attic and west to FAU-1, with a shutoff and sediment trap at the unit. Size it for the furnace, range and water heater (about 175 CFH at 7" w.c.: 3/4" to the mech room, 1/2" branches). The range branch GS5 tees off in the attic and drops inside wall 15 to a shutoff behind the range; the water heater branch GS4 stays in the mech room. CSST is bonded to the electrode system; the gas fitter does the pressure test.
6. Ventilation: EF2 in the primary bath runs continuously at 50 CFM to meet IRC M1505.4 whole-house ventilation, boosting to 80 for showers. An ERV is the better choice in a tight cold-climate house (open item).
7. Range hood 300 CFM over the gas range, ducted 6" to a roof cap; under 400 CFM, so no makeup air is required (M1503.6). Dryer vent rigid 4", 11' developed length with two elbows, roof cap on the front slope.

Open items

The particulars to settle on this sheet.

- Furnace in the attic: in a cold climate, the condensate lines need protection from freezing (route inside the insulation or add heat tape). Confirm, or move FAU-1 into a conditioned closet (the back hall alcove fits an upflow unit).
- Water heater: 50 gal power direct vent tank drawn. A gas tankless (about $1,500 more) saves the stand and space but needs a larger gas line.
- Whole-house ventilation: continuous bath fan (drawn) or an ERV (about $2,500–$4,000 installed)?
- Manual J/S/D from the HVAC contractor before ordering equipment; the 2.5-ton size is an estimate for 1,800 sq ft.

Equipment schedule

Tag	Equipment	Where	Size	Notes
FAU-1	Gas furnace, condensing, with A/C coil	attic over the back hall	60,000 BTU/h input, 96% AFUE, two-stage; 1,000 CFM variable speed; 2.5-ton coil	FAU-1: 96% AFUE two-stage condensing gas furnace, 60,000 BTU/h input, variable-speed blower, with a 2.5-ton cased A/C coil; listed for attic installation, horizontal on a platform over the back hall; 30" service platform at the access side, auxiliary drain pan with a float switch
AC-1	Air conditioner, outdoor	east side, by the mech room	2.5 ton, 15.2 SEER2 or better	AC-1: 2.5-ton air conditioner (condensing unit) on a pad, 18" off the east wall
GM	Gas meter	garage east wall	by the utility	Gas meter and regulator on the garage east wall, set by the utility; regulator vent 3' from the AC-1 unit, openings and the dryer exhaust
V1	Furnace vent	rear roof slope	2" PVC, concentric	FAU-1 vent and combustion air: 2" PVC intake and exhaust (direct vent, sealed combustion), up through the rear roof slope with a concentric termination 12" above the roof and the snow line; slope back to the furnace

Tag	Equipment	Where	Size	Notes
V2	Water heater vent	rear roof slope, over the mech room	PVC per the manual, concentric	Water heater vent and combustion air: 2" or 3" PVC per the manual (power direct vent), up through the garage ceiling and the rear roof slope, concentric termination 12" above the roof and the snow line
T1	Thermostat	wall 5, living room	communicating, with outdoor sensor	thermostat on wall 5, living-room side, 54" AFF; the hall return and the cased opening keep it in the house air
EF1	Bath 2 fan		80 CFM	4" duct to a roof cap, rear slope
EF2	Primary bath fan		80 CFM	4" duct to a roof cap, rear slope
EF3	Range hood		300 CFM	6" duct to a roof cap with damper, rear slope
EF4	Dryer vent		—	4" duct to a dryer roof cap, front slope; rigid metal, 11' developed with 2 elbows (M1502.4.5)

Register schedule

1000 CFM supply in all.

Tag	Room	CFM	Flex	Diffuser	Position
G1	Bedroom 3	100	7"	10×6 4-way ceiling diffuser	5'-7" from west ext. wall, 2'-11" from W6
G2	Bath 2	50	6"	8×6 ceiling diffuser	3'-2" from west ext. wall, 4'-3" from W7
G3	Bedroom 2	100	7"	10×6 4-way ceiling diffuser	5'-7" from west ext. wall, 2'-2" from rear ext. wall
G4	Living Room	100	7"	10×6 4-way ceiling diffuser	3'-2" from W5, 2'-4" from front ext. wall
G5	Living Room	100	7"	10×6 4-way ceiling diffuser	5'-10" from W11, 2'-4" from front ext. wall
G6	Dining	120	8"	10×6 4-way ceiling diffuser	5'-4" from W5, 1'-11" from rear ext. wall
G7	Kitchen	130	8"	10×6 4-way ceiling diffuser	1'-10" from W18, 4'-2" from rear ext. wall
G8	Mud / Laundry	50	6"	8×6 ceiling diffuser	1'-4" from W24, 3'-11" from front ext. wall
G9	Primary Bedroom	80	7"	10×6 4-way ceiling diffuser	4'-10" from W15, 1'-9" from rear ext. wall
G10	Primary Bedroom	70	6"	10×6 4-way ceiling diffuser	4'-10" from W15, 2'-1" from W14
G11	Primary Bath	70	6"	10×6 4-way ceiling diffuser	3'-5" from W16, 3'-3" from W14
G12	WIC	30	5"	8×6 ceiling diffuser	4'-3" from W16, 2'-4" from rear ext. wall

Return schedule

Tag	Room	CFM	Size	Notes
RG1	Back Hall	650	20×30 filter grille, MERV 8	central return under FAU-1, straight into the return plenum
RG2	Hall	350	16×25 filter grille, MERV 8	bedroom-wing return, 14" insulated flex to the return plenum
J1	Bedroom 3	—	10" jump duct	ceiling grille to the hall, so the room returns with its door shut
J2	Bedroom 2	—	10" jump duct	ceiling grille to the hall, so the room returns with its door shut
J3	Primary Bedroom	—	10" jump duct	ceiling grille to the hall, so the room returns with its door shut

Duct schedule

Trunk	Size	Length	Notes
ST1	14×8	18'-6"	west trunk in the attic, sheet metal, R-8 wrap; reduces to 10×8 past the living room take-offs
ST1b	10×8	12'-0"	bedroom-wing trunk over the hall

Trunk	Size	Length	Notes
ST2	14×8	24'-9"	north trunk over the kitchen
ST2b	10×8	14'-0"	primary-suite trunk

Bill of materials

Rough-order quantities from the drawing geometry, for pricing. The HVAC contractor takes off the final list.

Item	Qty	Notes
5" insulated flex, R-8	1 × 25' bags	10' measured +20%
6" insulated flex, R-8	2 × 25' bags	37' measured +20%
7" insulated flex, R-8	3 × 25' bags	61' measured +20%
8" insulated flex, R-8	2 × 25' bags	29' measured +20%
Sheet metal trunk 14×8 / 10×8, R-8 wrap	77'	plus take-off collars with dampers, one per register
Ceiling diffusers	12	with boots, sealed to the drywall
Filter return grilles 20×30 and 16×25	2	MERV 8 filters
Jump ducts, 10" flex with 2 grilles each	3	
Bath fans 80 CFM with roof caps	2	EF2 with a continuous-ventilation control
Gas piping: 3/4" black steel or CSST, 1/2" CSST	60'	shutoffs, sediment traps, CSST bonding clamp and #6 Cu; pressure test 3 psi for 10 min (G2417)
2" PVC sch 40 intake and exhaust, concentric roof kit	1 kit + 20'	per the furnace manual
Mastic, UL 181 tape, duct straps	lot	every joint sealed; total leakage ≤ 4 CFM25 per 100 sq ft (IECC R403.3.5)